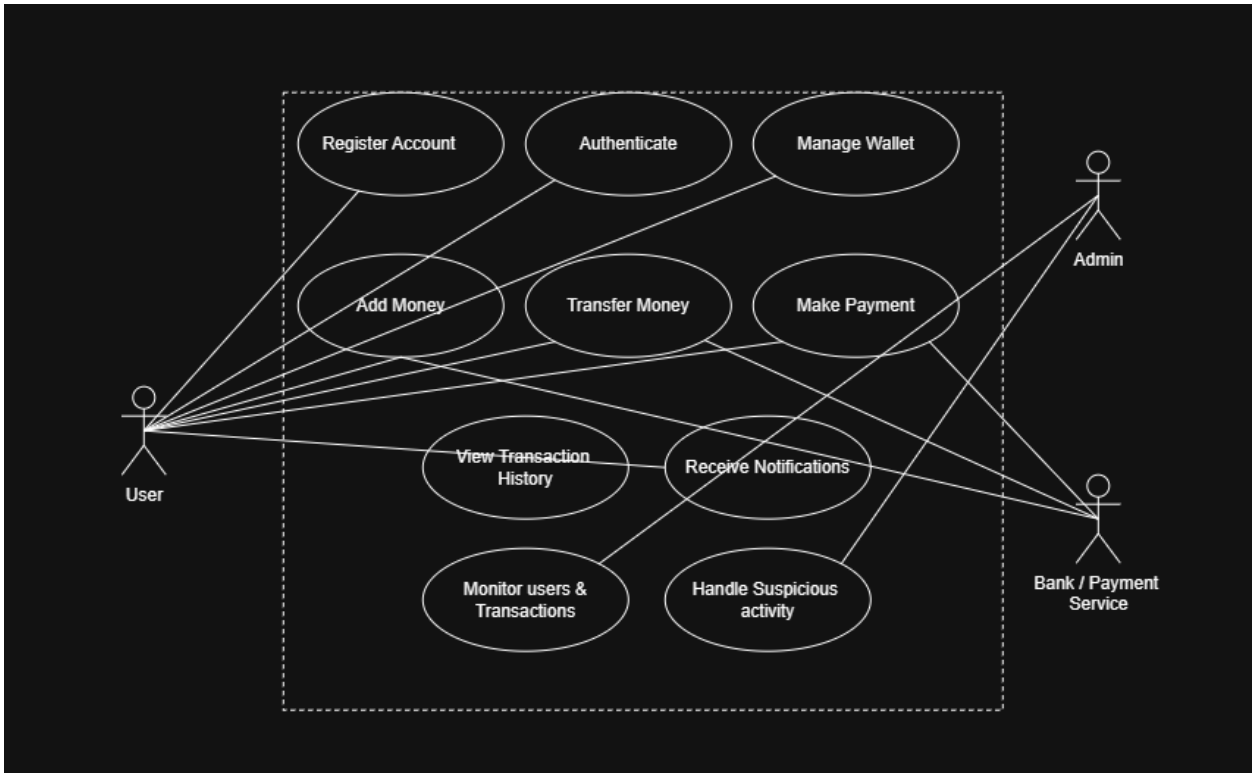
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## 1. Use Case Diagram




## 2. Actors and Their Roles

**User / Customer:** The primary user class. Registers, authenticates, and performs wallet management, money transfers, payments, and transaction history operations.

**Administrator:** An internal, trusted user with elevated privileges who monitors users and transactions and investigates/handles suspicious activity.

**Bank / Payment Service:** An external system/gateway that validates and processes supported financial operations (funding, transfers, payments) on the DWMS's behalf.

## 3. Use Case Specifications

### Register Account

**Function:** Allows a new user to create and verify a digital wallet account.

**Precondition:** The user does not have an existing active account.

**Postcondition:** A new account is created, verified, and ready for use.

### Authenticate

**Function:** Verifies a user's credentials securely before granting access to wallet functions.

**Precondition:** The user has a registered and verified account.

**Postcondition:** The user is successfully logged in and granted access to their wallet dashboard.

### Manage Wallet

**Function:** Allows an authenticated user to view their wallet balance and manage wallet settings.

**Precondition:** The user is successfully authenticated.

**Postcondition:** The current wallet balance and related settings are displayed to the user.

### Add Money

**Function:** Allows a user to add funds to their wallet through a supported external funding method.

**Precondition:** The user is authenticated and selects a valid funding method.

**Postcondition:** The wallet balance is updated, and the funding transaction is recorded in the ledger upon confirmation.

### Transfer Money

**Function:** Enables an authenticated user to send money to another registered user of the system.

**Precondition:** The user is authenticated, has a sufficient wallet balance, and the recipient account exists.

**Postcondition:** Both the sender's and recipient's wallet balances are updated atomically, and the transfer is recorded.

### Make Payment

**Function:** Allows an authenticated user to initiate and confirm a payment to a merchant or third-party payee.

**Precondition:** The user is authenticated and has sufficient balance for the payment amount.

**Postcondition:** The payment is processed via the payment gateway, the ledger is updated, and the final status is reported.

### View Transaction History

**Function:** Allows a user to view, filter, and review details of their past transactions.

**Precondition:** The user is authenticated.

**Postcondition:** A list of past transactions matching the specified filters is displayed to the user.

### Receive Notifications

**Function:** Delivers alerts and notifications to users regarding transaction status and significant account events.

**Precondition:** A qualifying account event or transaction completion occurs.

**Postcondition:** A notification is generated and delivered to the user via their configured communication channel.

### Monitor Users & Transactions

**Function:** Allows administrators to view user accounts and search/monitor transactions across the entire system.

**Precondition:** The administrator is authenticated with elevated administrative privileges.

**Postcondition:** The administrator is presented with user and transaction records matching their criteria.

### Handle Suspicious Activity

**Function:** Enables administrators to flag or restrict accounts suspected of fraudulent or suspicious behavior.

**Precondition:** The administrator identifies anomalous activity during monitoring.

**Postcondition:** The suspicious account is restricted or flagged, and the action is recorded in the audit log.